

Practical No.2

Aim: Write an R program to define and print different types of data structures commonly used in R. Your program should include the following:

Theory:

R is a programming language used for statistical computing and data analysis. It provides various basic data types and data structures to store and manipulate data efficiently.

1. Basic Data Types in R:

- Numeric: Represents decimal numbers (e.g., 10.5)
- Integer: Represents whole numbers (e.g., 10L)
- Character: Represents text or strings (e.g., "Hello")
- Logical: Represents Boolean values (TRUE/FALSE)

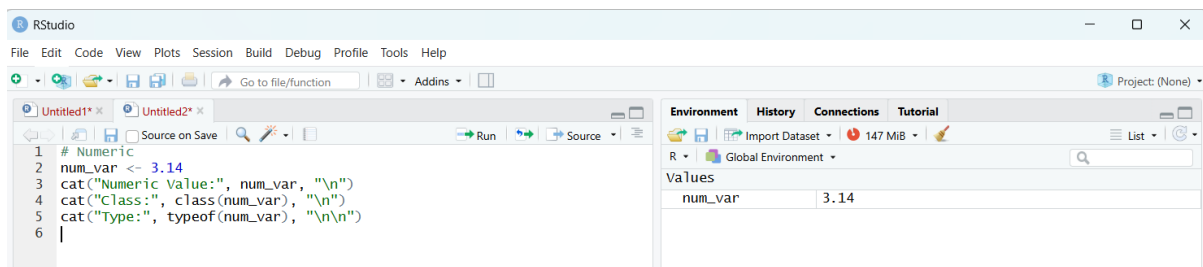
2. Data Structures in R:

- Vector: Collection of elements of the same data type
- List: Collection of elements of different data types
- Matrix: Two-dimensional homogeneous data structure
- Data Frame: Tabular structure with rows and columns, can hold different data types

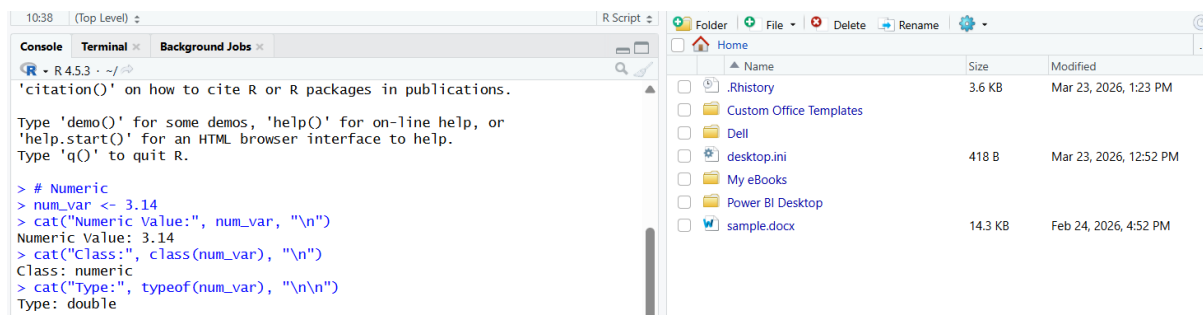
3. Functions Used:

- `print()` → Displays output
- `cat()` → Prints formatted output
- `class()` → Returns the class of an object
- `typeof()` → Returns the internal storage type

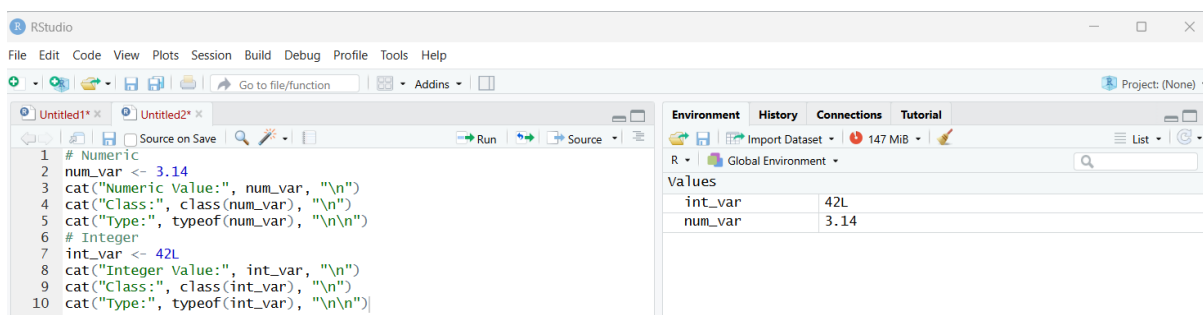
Code for numeric:



Output:



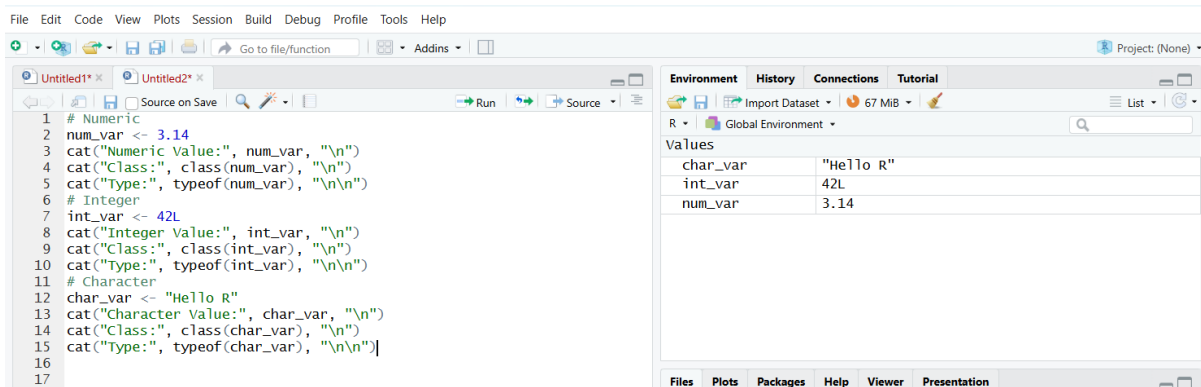
Code integer:



Output for integer:

```
> # Integer
> int_var <- 42L
> cat("Integer Value:", int_var, "\n")
Integer Value: 42
> cat("Class:", class(int_var), "\n")
Class: integer
> cat("Type:", typeof(int_var), "\n\n")
Type: integer
```

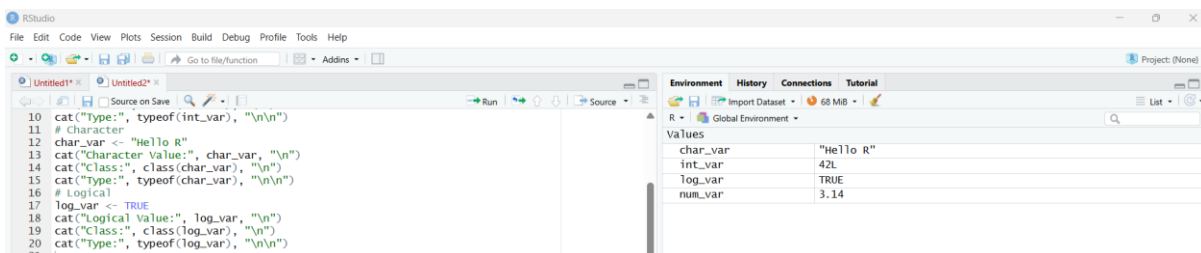
Code for character:



Output for character:

```
> # Character
> char_var <- "Hello R"
> cat("Character Value:", char_var, "\n")
Character Value: Hello R
> cat("Class:", class(char_var), "\n")
Class: character
> cat("Type:", typeof(char_var), "\n\n")
Type: character
```

Code for logical:

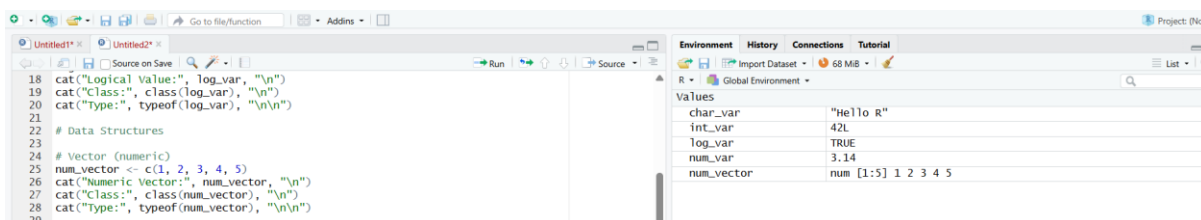


Output:

```
> # Logical
> log_var <- TRUE
> cat("Logical Value:", log_var, "\n")
Logical Value: TRUE
> cat("Class:", class(log_var), "\n")
Class: logical
> cat("Type:", typeof(log_var), "\n\n")
Type: logical

> |
```

Code for vector(numeric):



Output:

```

> # Data Structures
>
> # Vector (numeric)
> num_vector <- c(1, 2, 3, 4, 5)
> cat("Numeric Vector:", num_vector, "\n")
Numeric Vector: 1 2 3 4 5
> cat("Class:", class(num_vector), "\n")
Class: numeric
> cat("Type:", typeof(num_vector), "\n\n")
Type: double
>

```

Code for vector(character)

The screenshot shows the RStudio interface. The script editor on the left contains the following code:

```

21 # Data Structures
22
23 # Vector (numeric)
24 num_vector <- c(1, 2, 3, 4, 5)
25 cat("Numeric Vector:", num_vector, "\n")
26 cat("Class:", class(num_vector), "\n")
27 cat("Type:", typeof(num_vector), "\n\n")
28
29 # Vector (character)
30 char_vector <- c("apple", "banana", "cherry")
31 cat("Character Vector:", char_vector, "\n")
32 cat("Class:", class(char_vector), "\n")
33 cat("Type:", typeof(char_vector), "\n\n")
34

```

The Environment pane on the right shows the following variables:

Variable	Value
char_var	"Hello R"
char_vector	chr [1:3] "apple" "banana" "cherry"
int_var	42L
log_var	TRUE
num_var	3.14
num_vector	num [1:5] 1 2 3 4 5

Output:

```

> # Vector (character)
> char_vector <- c("apple", "banana", "cherry")
> cat("Character Vector:", char_vector, "\n")
Character Vector: apple banana cherry
> cat("Class:", class(char_vector), "\n")
Class: character
> cat("Type:", typeof(char_vector), "\n\n")
Type: character
>

```

Code for list(mixed datatypes):

The screenshot shows the RStudio interface. The script editor on the left contains the following code:

```

28 cat("Type:", typeof(num_vector), "\n\n")
29
30 # Vector (character)
31 char_vector <- c("apple", "banana", "cherry")
32 cat("Character Vector:", char_vector, "\n")
33 cat("Class:", class(char_vector), "\n")
34 cat("Type:", typeof(char_vector), "\n\n")
35
36 # List (mixed data types)
37 mixed_list <- list(123, "R programming", TRUE, c(1,2,3))
38 print(mixed_list)
39 cat("Class:", class(mixed_list), "\n")
40 cat("Type:", typeof(mixed_list), "\n\n")
41

```

The Environment pane on the right shows the following variables:

Variable	Value
mixed_list	List of 4
\$	num 123
\$	chr "R programming"
\$	logi TRUE
\$	num [1:3] 1 2 3
char_var	"Hello R"
char_vector	chr [1:3] "apple" "banana" "cherry"
int_var	42L

Output:

```
40:26 (Top Level) R Script

Console Terminal Background Jobs

R 4.5.3 ~ /

> # List (mixed data types)
> mixed_list <- list(123, "R programming", TRUE, c(1,2,3))
> cat("List:\n")
List:
> print(mixed_list)
[[1]]
[1] 123

[[2]]
[1] "R programming"

[[3]]
[1] TRUE

[[4]]
[1] 1 2 3

> cat("Class:", class(mixed_list), "\n")
Class: list
> cat("Type:", typeof(mixed_list), "\n\n")
Type: list
```

Code for matrix:

```
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1* Untitled2*
Source on Save Run Go to file/function Addins
35
36 # List (mixed data types)
37 mixed_list <- list(123, "R programming", TRUE, c(1,2,3))
38 cat("List:\n")
39 print(mixed_list)
40 cat("Class:", class(mixed_list), "\n")
41 cat("Type:", typeof(mixed_list), "\n\n")
42
43 # Matrix
44 mat <- matrix(1:9, nrow=3, ncol=3)
45 cat("Matrix:\n")
46 print(mat)
47 cat("Class:", class(mat), "\n")
48 cat("Type:", typeof(mat), "\n\n")
49
40:39 (Top Level) R Script

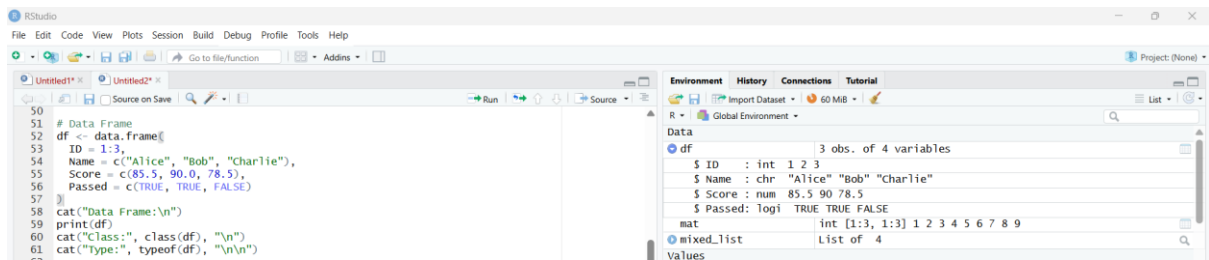
Environment History Connections Tutorial
R Global Environment
Data
mat int [1:3, 1:3] 1 2 3 4 5 6 7 8 9
mixed_list List of 4
Values
char_var "Hello R"
char_vector chr [1:3] "apple" "banana" "cherry"
int_var 42L
log_var TRUE
num_var 3.14
num_vector num [1:5] 1 2 3 4 5
Files Plots Packages Help Viewer Presentation
New Folder New File Delete Rename More
```

Output:

```
> # Matrix
> mat <- matrix(1:9, nrow=3, ncol=3)
> cat("Matrix:\n")
Matrix:
> print(mat)
[,1] [,2] [,3]
[1,] 1 4 7
[2,] 2 5 8
[3,] 3 6 9
> cat("Class:", class(mat), "\n")
Class: matrix array
> cat("Type:", typeof(mat), "\n\n")
Type: integer

> |
```

Code for dataframe:



Output:

```
> # Data Frame
> df <- data.frame(
+   ID = 1:3,
+   Name = c("Alice", "Bob", "Charlie"),
+   Score = c(85.5, 90.0, 78.5),
+   Passed = c(TRUE, TRUE, FALSE)
+ )
> cat("Data Frame:\n")
Data Frame:
> print(df)
  ID Name Score Passed
1  1 Alice  85.5   TRUE
2  2  Bob  90.0   TRUE
3  3 Charlie 78.5  FALSE
> cat("Class:", class(df), "\n")
Class: data.frame
> cat("Type:", typeof(df), "\n\n")
Type: list
> |
```

Conclusion: Thus, different basic data types (numeric, integer, character, logical) and data structures (vector, list, matrix, data frame) in R were successfully created and displayed along with their values, classes, and internal data types.